MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
Even Semester Examinations 2023-24
Paper Code: PEC-CS601B
Paper Name: Distributed Database

Time Allotted : **3 Hours** *Full Marks :* **70**

The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable.

Answer all the questions

GROUP – A(Multiple Choice Type Questions)

1	**Choose the correct alternatives of the following :**	**10 X 1 = 10**		
		MARKS	***Mapped CO***	***Bloom's Level***
i)	A distributed database has which of the following advantages over centralized database ?	1	CO1	BL-1
a.	Software cost			
b.	Software complexity			
c.	Slow response			
d.	Modular growth			
ii)	An autonomous homogenous environment is which of the following ?	1	CO1	BL-1
a.	The same DBMS is at each node and each DBMS works independently.			
b.	The same DBMS is at each node and a central DBMS coordinates database access.			
c.	A different DBMS is at each node and each DBMS works independently			
d.	A different DBMS is at each node and a central DBMS coordinates database access			
iii)	Location transparency allows for which of the following ?	1	CO1	BL-1
a.	Users to treat the data as if it is at one location			
b.	Programmers to treat the data as if it is at one location			
c.	Managers to treat the data as if it is at one location			
d.	All of the above			
iv)	Which of the following is true concerning a global transaction?	1	CO1	BL-1
a.	The required data are at one local site and the distributed DBMS routes requests as necessary			
b.	The required data are located in at least one non-local site and the distributed DBMS routes requests as necessary.			
c.	The required data are at one local site and the distributed DBMS passes the request to only the local DBMS.			
d.	The required data are located in at least one non-local site and the distributed DBMS passes the request to only the local DBMS.			
v)	Storing a separate copy of the database at multiple locations is which of the following ?	1	CO1	BL-1
a.	Data replication			
b.	Data sharing			

c.	Data mirroring			
d.	Data fragmentation			
vi)	Which of the following is an objective of data distribution in the distributed database system?	1	CO1	BL-1
a.	Locality of reference			
b.	Availability and reliability			
c.	Communication cost			
d.	All of these			
vii)	Which transparency is provided if users are unaware about the location of the data objects ?	1	CO1	BL-1
a.	Naming transparency			
b.	Location transparency			
c.	Replication transparency			
d.	Fragmentation transparency			
viii)	___ transparency exists when the end user or programmer must specify both the fragment names and their locations.	1	CO1	BL-1
a.	Local mapping			
b.	Replication			
c.	Naming			
d.	None of these			
ix)	Some of the columns of a relation are at different sites is which of the following?	1	CO1	BL-1
a.	Data replication			
b.	Horizontal partitioning			
c.	Vertical partitioning			
d.	None of these			
x)	A fragmentation technique where in every tuple of a table is assigned to one or more fragment as a result of fragmentation is called	1	CO1	BL-1
a.	Vertical fragmentation			
b.	Horizontal fragmentation			
c.	Hybrid fragmentation			
d.	None of these			
	GROUP – B(Short Answer Type Questions)		**3 X5 = 15**	
	Answer the following.	**MARKS**	***Mapped CO***	***Bloom's Level***
2.a.	Define distributed database. List the advantages of it.	2+3	CO1	BL-1
OR 2.b.	Describe the objectives of data distribution.	5	CO1	BL-1
3.a.	What is Distributed database? How it differs from Centralized Database ?	5	CO1	BL-1
OR 3.b.	What are homogenous and heterogeneous database.	5	CO1	BL-1
4.a.	Explain canonical form of global query with suitable example.	5	CO2	BL-2
OR 4.b.	Differentiate Operator tree and Optimization graph with suitable example.	5	CO2	BL-2
	GROUP – C(Long Answer Type Questions)		**3 X15 = 45**	
	Answer the following.	**MARKS**	***Mapped CO***	***Bloom's Level***
5.a.i.	Explain the reference architecture of DDB with a diagram.	10	CO1	BL-2
ii.	"Bottom-up approach of distributed database design is applicable for integrating existing databases"; justify your answer.	5	CO1	BL-2

OR 5.b.i.	What do you mean by fragmented schema? What are the advantages of keeping fragmentation schema independent of allocation schema?	2 + 3	CO1	BL-2
ii.	Write about the completeness, reconstruction and disjointness condition rules for defining fragmentation.	6	CO1	BL-2
iii	What are the advantages and disadvantages of replication?	4	CO1	BL-2
6.a.i.	Define transaction. Discuss about the different type of agents in distributed transaction.	1+5	CO3	BL-1
ii.	Write about the atomicity of transactions in distributed databases.	5	CO3	BL-2
iii.	What is network partitioning?	2	CO3	BL-1
iv	Define K-resilient system with respect to site failures	2	CO3	BL-1
OR 6.b.i.	Define distributed serializability. How is it ensured in DDBMS?	2+3	CO3	BL-2
ii.	Differentiate between basic TO algorithm and conservative TO algorithm.	6	CO3	BL-2
iii.	How disadvantages of 2PL are overcome in strict 2PL ?	4	CO3	BL-3
7.a.i.	What is wait for graph? Differentiate between Local wait for graph (LWFG) and Global wait for graph (DWFG). Explain the use of DWFG with example.	2+3+5	CO4	BL-2
ii.	Explain false deadlock with example.	5	CO4	BL-2
OR 7.b.i.	Differentiate between pre-emptive and non-preemptive methods for distributed deadlock prevention.	5	CO4	BL-2
ii.	How multiple copies of data can be handled by 2-phase locking protocol as a concurrency control mechanism in distributed database?	5	CO4	BL-2
iii.	Compare pessimistic and optimistic concurrency control techniques.	5	CO4	BL-2